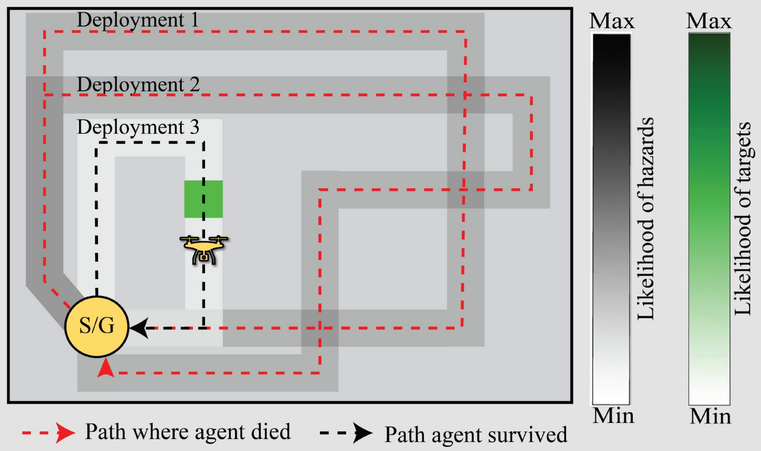
Estimating Hazardous Locations in Communication-Denied Environments: Distributed Multi-Robot Approaches with Path-Based Sensors

Alkesh K. Srivastava[1], George P. Kontoudis[1], Donald Sofge[2], Michael Otte[1]
[1]Motion and Teaming Lab, University of Maryland, College Park [2]U.S. Naval Research Laboratory, Washington, D.C

MOTIVATION



- In a communication-deprived environment, a hazard may ***detain***, ***disable***, or ***destroy*** the robot
- Such situations are common in disaster-stricken ***search & rescue missions, military operations*** in hostile environments, and ***under -water exploration*** and ***surveying***
- In such environments, ***secure*** and ***effective*** information gathering about targets and hazards is crucial
- ***Distributed multi-agent*** approaches can improve the ***efficiency, security,*** and ***robustness*** of information gathering

RESEARCH OBJECTIVE

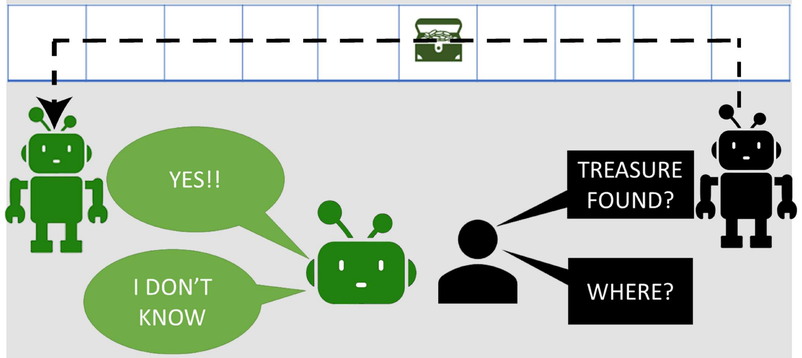
- Given a team of ***M*** agents with ***communication forbidden*** among them, the objective of agents is to efficiently and robustly ***gather information*** about the targets ***X*** and hazards ***Z***
- Develop ***distributed methodologies*** for information gathering in a ***multi-robot system*** using ***path-based sensors***
- Reduce the ***computational complexity*** of the information

PROPOSED METHODS

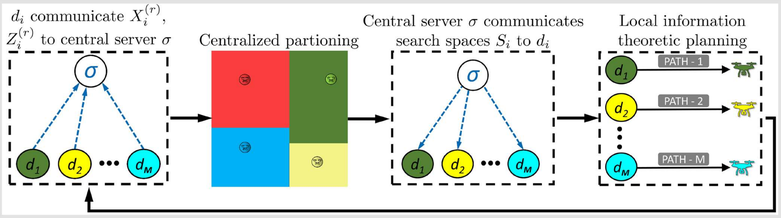
- **Distributed Entropy Voronoi Partition and Planner (DEVPP)**
  Improved Effifacy ☑ Distributed ☑ Reduced Computation ☑
- **Multi-agent Distributed Information-theoretic Planner (MA-DITP)**
  Improved Effifacy ☑ Distributed ☑ Reduced Computation ☐
- **Multi-agent Global Information-theoretic Planner (MA-GITP)**
  Improved Effifacy ☑ Distributed ☐ Reduced Computation ☐

PATH-BASED SENSOR

A sensor that reports whether or not an event has occurred somewhere along a path, but has ***no conception*** of where along the path that event has taken place.



DEVPP



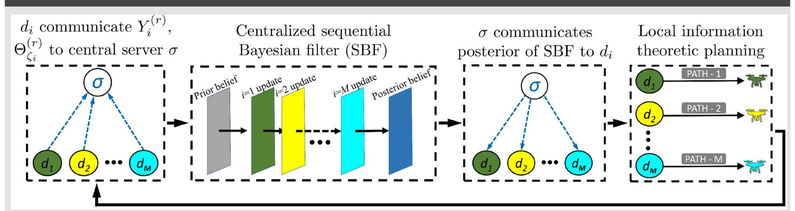
Step 1: Centralized Entropy Decomposition:

$$S_i = \{\, \boldsymbol{p}_i \in \mathcal{S} \mid w(X_{\boldsymbol{p}_i}, X_{S_j}) g(\boldsymbol{p}_i, \boldsymbol{p}_{d_j}) \leq w(X_{\boldsymbol{p}_i}, X_{S_k}) g(\boldsymbol{p}_i, \boldsymbol{p}_{d_k}),\ \forall\, j \neq k \}$$

Step 2: Local Information-theoretic Planning:

$$\zeta_i^{(r)*} = \arg\max_{\zeta_i \in \Omega_i} \left\{ I(X_i^{(r-1)};\ \boldsymbol{Y}_i^{(r)} \mid \Theta_{\zeta_i}^{(r)}, S_i^{(r)}) + I(Z_i^{(r-1)}; \Theta_{\zeta_i}^{(r)} \mid S_i^{(r)}) \right\}$$

MA-DITP

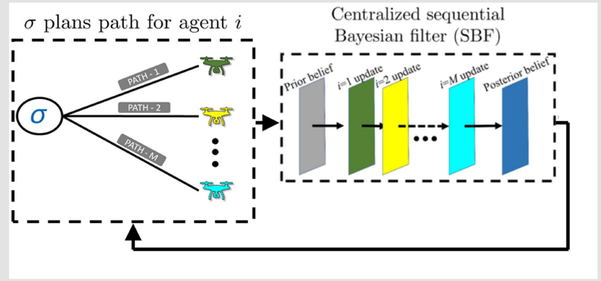


- Each agent plans a path in a ***distributed manner*** using the information-theoretic planner:

$$\zeta_i^{(r)*} = \arg\max_{\zeta_i \in \Omega_i} \left\{ I(X_i^{(r-1)};\ \boldsymbol{Y}_i^{(r)} \mid \Theta_{\zeta_i}^{(r)}, S_i^{(r)}) + I(Z_i^{(r-1)}; \Theta_{\zeta_i}^{(r)} \mid S_i^{(r)}) \right\}$$

- A ***Sequential Bayesian Filter*** is used to update belief of the environment

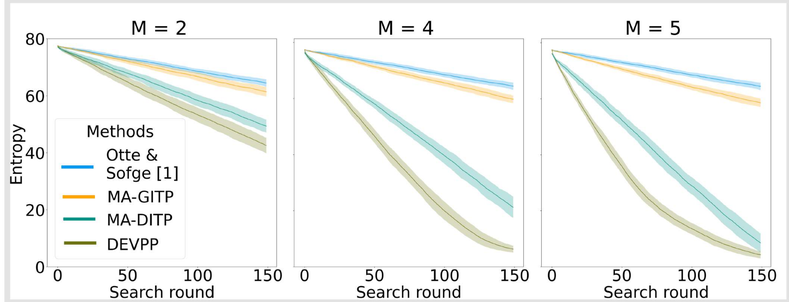
MA-GITP



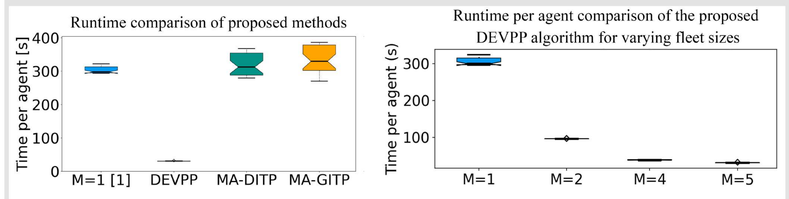
- ***Centralized server*** plans paths for agents using the information-theoretic planner
- A ***Sequential Bayesian Filter*** is used to update belief map of the environment
- Ideal for ***secruity critical*** missions

RESULTS

- **DEVPP *outperforms*** all other proposed algorithms in terms of ***information gathering***



- **DEVPP** has the ***minimal run-time*** per agent



REFERENCES

[1] Otte, Michael, and Donald Sofge. "Path-based sensors: Paths as Sensors, Bayesian updates, and Shannon information gathering." IEEE Transactions on Automation Science and Engineering 18.3 (2021): 946-967.

[2] Srivastava, Alkesh K., et al. "Distributed Multi-Robot Information Gathering using Path-Based Sensors in Entropy-Weighted Voronoi Regions." International Symposium on Distributed Autonmous Robotic Systems (DARS), Montbéliard, France. 2022.